



Université Internationale de Rabat (UIR)

Big Data & Artificial Intelligence

Smart Temperature Alarm System

IoT Embedded Systems Project using Arduino Uno & DHT11

Prepared by:

- Manal HOUMAILY
- Oumaima GASSI
- Salma BENMOUMA
- Mohamed Taha El Qassid

Academic Year:

2025 – 2026

Acknowledgements

We would like to express our sincere gratitude to the **Université Internationale de Rabat (UIR)** and the Big Data & Artificial Intelligence program for providing an excellent academic environment that encourages innovation, practical learning, and hands-on embedded systems projects.

Contents

Acknowledgements	1
Project Images	4
1 General Introduction	6
2 Hardware Components	7
2.1 Overview	7
2.2 Arduino Uno	7
2.3 DHT11 Sensor	7
2.4 LCD 16x2 with I2C	8
2.5 LEDs and Buzzer	8
3 Circuit and Pin Connections	9
3.1 Wiring Table	9
3.2 Circuit Schematic	9
4 System Algorithm and Logic	11
4.1 Operating Principle	11
4.2 Logic Flow	11
4.3 Threshold Configuration	11
5 Arduino Source Code	12
5.1 Full Sketch	12
5.2 Required Libraries	15
6 Results and Testing	16
6.1 System Behavior	16
6.2 Serial Monitor Output	16
6.3 Issues and Solutions	17
7 General Conclusion	18

List of Figures

1	All Components — Arduino Uno, DHT11, Buzzer, LEDs, Wires	4
2	System-Logic-flow	4
3	System Running — Serial Monitor Dashboard	5

Project Images

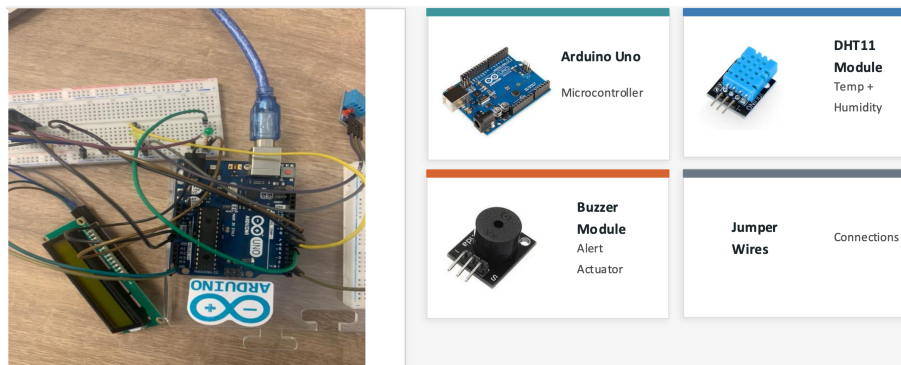


Figure 1: All Components — Arduino Uno, DHT11, Buzzer, LEDs, Wires

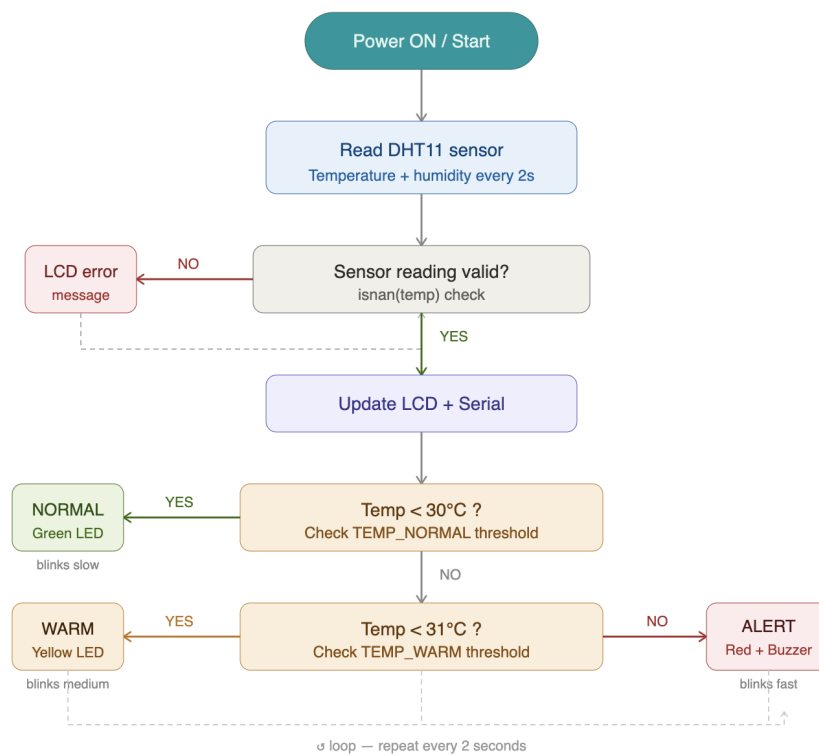


Figure 2: System-Logic-flow

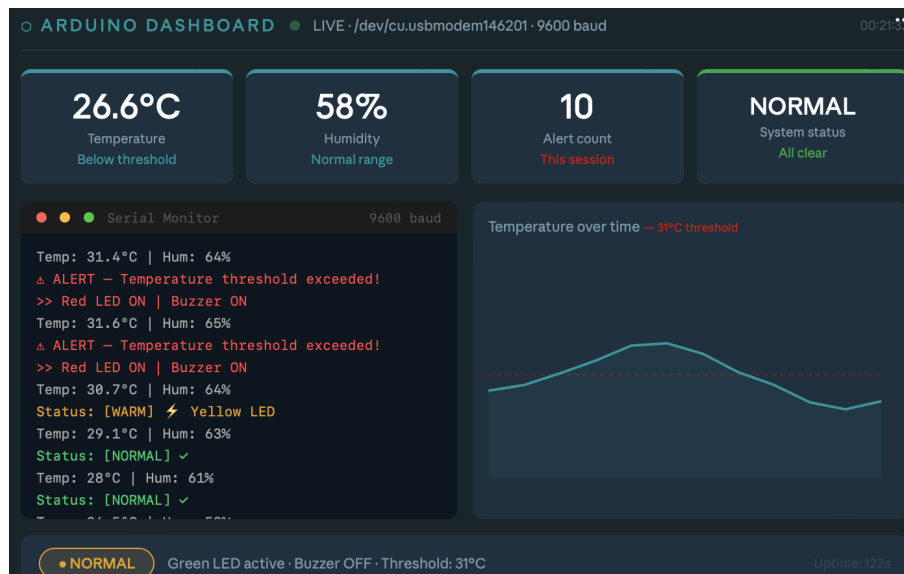


Figure 3: System Running — Serial Monitor Dashboard

Chapter 1

General Introduction

The Internet of Things (IoT) has revolutionized the way we interact with the physical world. Embedded systems based on microcontrollers such as the **Arduino Uno** enable the development of smart, real-time monitoring systems at low cost and minimal complexity.

This project presents a **Smart Temperature Alarm System** that continuously monitors ambient temperature and humidity using a **DHT11 sensor**. Based on configurable thresholds, the system activates visual indicators (LEDs) and an audible alert (buzzer) to notify users of abnormal temperature conditions.

The system implements three operating states:

- **Normal** (below 30°C) — Green LED blinks slowly, buzzer silent.
- **Warm** (30°C to 31°C) — Yellow LED blinks at medium speed, buzzer silent.
- **Alert** (above 31°C) — Red LED blinks rapidly, buzzer activated.

Chapter 2

Hardware Components

2.1 Overview

Component	Role	Qty	Notes
Arduino Uno	Microcontroller	1	ATmega328P
DHT11 Module	Temp + Humidity	1	Digital sensor
LCD 16x2 I2C	Display	1	Address 0x27
Green LED	Normal indicator	1	220 Ω resistor
Yellow LED	Warm indicator	1	220 Ω resistor
Red LED	Alert indicator	1	220 Ω resistor
Piezo Buzzer	Sound alert	1	Active buzzer
Breadboard	Connections	1	Standard
Jumper Wires	Wiring	–	Male-to-Male

Table 2.1: Hardware Components List

2.2 Arduino Uno

The **Arduino Uno** is the central processing unit. It reads sensor data, applies threshold logic, and controls all output components. It operates at 5V with 14 digital I/O pins and 6 analog pins.

2.3 DHT11 Sensor

The **DHT11** measures:

- Temperature: 0°C to 50°C ($\pm 2^\circ\text{C}$ accuracy)
- Humidity: 20% to 80% ($\pm 5\%$ accuracy)

- Sampling rate: one reading every 2 seconds

2.4 LCD 16x2 with I2C

Displays real-time temperature, humidity, and system status. The I2C module reduces wiring to only 2 data pins (SDA/SCL).

2.5 LEDs and Buzzer

Three LEDs provide immediate visual feedback. The active piezo buzzer emits a beeping alert when temperature exceeds the high threshold.

Chapter 3

Circuit and Pin Connections

3.1 Wiring Table

Component	Pin	Arduino Pin
DHT11	DATA	Digital 2
DHT11	VCC	5V
DHT11	GND	GND
Green LED	+ via 220 Ω	Digital 6
Yellow LED	+ via 220 Ω	Digital 7
Red LED	+ via 220 Ω	Digital 8
Buzzer	+ (positive)	Digital 9
Buzzer	- (negative)	GND
LCD	SDA	A4
LCD	SCL	A5
LCD	VCC	5V
LCD	GND	GND

Table 3.1: Pin Connection Table

3.2 Circuit Schematic

```
1 Arduino Uno
2 +-----+
3 | D2  ----- DHT11 DATA  |
4 | D6  ---  2200hm  ---  Green LED  |
5 | D7  ---  2200hm  ---  Yellow LED |
6 | D8  ---  2200hm  ---  Red LED   |
7 | D9  ----- BUZZER (+)   |
```

8		A4	-----	LCD	SDA	
9		A5	-----	LCD	SCL	
10		5V	-----	VCC	(all)	
11		GND	-----	GND	(all)	
12	+-----+					

Chapter 4

System Algorithm and Logic

4.1 Operating Principle

The system operates in a continuous loop, reading sensor data every 2 seconds and applying threshold-based logic.

4.2 Logic Flow

1. **Power ON** — Initialize Serial, DHT11, LCD, pins.
2. **Read Sensor** — Get temperature and humidity.
3. **Validate** — Check for errors with `isnan()`.
4. **Update Display** — Show values on LCD and Serial Monitor.
5. **Apply Logic:**
 - $\text{temp} < 30^{\circ}\text{C} \rightarrow \text{NORMAL}$ — Green LED, buzzer OFF
 - $30^{\circ}\text{C} \leq \text{temp} < 31^{\circ}\text{C} \rightarrow \text{WARM}$ — Yellow LED, buzzer OFF
 - $\text{temp} \geq 31^{\circ}\text{C} \rightarrow \text{ALERT}$ — Red LED, buzzer ON
6. **Repeat** — Wait 2 seconds, loop again.

4.3 Threshold Configuration

```
1 #define TEMP_NORMAL 30.0 // Normal -> Warm (Celsius)
2 #define TEMP_WARM 31.0 // Warm -> Alert (Celsius)
```

Chapter 5

Arduino Source Code

5.1 Full Sketch

```
1  #include <DHT.h>
2  #include <LiquidCrystal_I2C.h>
3
4  #define DHTPIN      2
5  #define DHTTYPE     DHT11
6  #define GREENLED    6
7  #define YELLOWLED   7
8  #define REDLED      8
9  #define BUZZER      9
10 #define TEMP_NORMAL 30.0
11 #define TEMP_WARM   31.0
12
13 DHT dht(DHTPIN, DHTTYPE);
14 LiquidCrystal_I2C lcd(0x27, 16, 2);
15
16 void allLedsOff() {
17     digitalWrite(GREENLED, LOW);
18     digitalWrite(YELLOWLED, LOW);
19     digitalWrite(REDLED, LOW);
20     noTone(BUZZER);
21     digitalWrite(BUZZER, LOW);
22 }
23
24 void setup() {
25     Serial.begin(9600);
26     dht.begin();
27     pinMode(GREENLED, OUTPUT);
```

```
28   pinMode(YELLOWLED, OUTPUT);
29   pinMode(REDLED,    OUTPUT);
30   pinMode(BUZZER,    OUTPUT);
31   allLedsOff();
32   lcd.init();
33   lcd.backlight();
34   lcd.setCursor(0, 0);
35   lcd.print("  Temp  Alarm  ");
36   lcd.setCursor(0, 1);
37   lcd.print("  Starting...  ");
38   delay(2000);
39   lcd.clear();
40 }
41
42 void loop() {
43   float temp = dht.readTemperature();
44   float hum  = dht.readHumidity();
45
46   if (isnan(temp) || isnan(hum)) {
47     allLedsOff();
48     lcd.setCursor(0, 0);
49     lcd.print("Sensor Error!  ");
50     lcd.setCursor(0, 1);
51     lcd.print("Check DHT11 !!  ");
52     Serial.println("ERROR: DHT11 failed");
53     delay(2000);
54     return;
55   }
56
57   lcd.setCursor(0, 0);
58   lcd.print("Temp: ");
59   lcd.print(temp, 1);
60   lcd.print((char)223);
61   lcd.print("C  ");
62   lcd.setCursor(8, 0);
63   lcd.print("H: ");
64   lcd.print(hum, 0);
65   lcd.print("%  ");
66
67   Serial.print("Temp: "); Serial.print(temp);
68   Serial.print("C | Hum: "); Serial.print(hum);
```

```
69 Serial.println("%");
70
71 if (temp < TEMP_NORMAL) {
72     allLedsOff();
73     digitalWrite(BUZZER, LOW);
74     lcd.setCursor(0, 1);
75     lcd.print("Status: NORMAL ");
76     Serial.println("Status: NORMAL");
77     digitalWrite(GREENLED, HIGH);
78     delay(900);
79     digitalWrite(GREENLED, LOW);
80     delay(100);
81 }
82 else if (temp >= TEMP_NORMAL && temp < TEMP_WARM) {
83     allLedsOff();
84     digitalWrite(BUZZER, LOW);
85     lcd.setCursor(0, 1);
86     lcd.print("Status: WARM ");
87     Serial.println("Status: WARM");
88     digitalWrite(YELLOWLED, HIGH);
89     delay(400);
90     digitalWrite(YELLOWLED, LOW);
91     delay(100);
92 }
93 else {
94     allLedsOff();
95     lcd.setCursor(0, 1);
96     lcd.print("!! HIGH ALERT !!");
97     Serial.println("Status: HIGH ALERT!!!");
98     for (int i = 0; i < 3; i++) {
99         digitalWrite(REDLED, HIGH);
100        digitalWrite(BUZZER, HIGH);
101        delay(200);
102        digitalWrite(REDLED, LOW);
103        digitalWrite(BUZZER, LOW);
104        delay(200);
105    }
106 }
107 }
```

5.2 Required Libraries

Library	Author	Purpose
DHT sensor library	Adafruit	DHT11 sensor
Adafruit Unified Sensor	Adafruit	Dependency
LiquidCrystal I2C	Frank de Brabander	LCD display

Table 5.1: Required Arduino Libraries

Chapter 6

Results and Testing

6.1 System Behavior

State	Condition	LED	Buzzer
NORMAL	Temp < 30°C	Green (slow)	OFF
WARM	30°C – 31°C	Yellow (medium)	OFF
ALERT	Temp ≥ 31°C	Red (fast)	ON

Table 6.1: System States and Outputs

6.2 Serial Monitor Output

```
1 Temp: 26.4C | Hum: 59%
2 Status: NORMAL
3 Temp: 30.8C | Hum: 63%
4 Status: WARM
5 Temp: 31.5C | Hum: 65%
6 Status: HIGH ALERT!!!
```

6.3 Issues and Solutions

Issue	Solution
Buzzer active at low temp	Added <code>digitalWrite(BUZZER, LOW)</code> in NORMAL and WARM states
Port not found on Mac	Used <code>ls /dev/cu.*</code> in Terminal
DHT library missing	Installed via Library Manager
LCD shows nothing	Changed I2C address to 0x3F

Table 6.2: Issues and Solutions

Chapter 7

General Conclusion

This project successfully demonstrates the implementation of a **Smart Temperature Alarm System** using Arduino Uno and DHT11 sensor. The system reliably monitors temperature and humidity in real time, providing immediate visual and audible feedback through three clearly defined operating states.

Key outcomes:

- Real-time sensor reading every 2 seconds.
- Three-level alert logic with configurable thresholds.
- LCD display showing live temperature and humidity.
- Serial Monitor dashboard for data visualization.
- Robust error handling for sensor failures.

This project lays the foundation for more advanced IoT applications such as smart home climate control, greenhouse monitoring, server room management, and industrial safety systems.